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Using DAX in Microsoft Power BI

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Your presenter today

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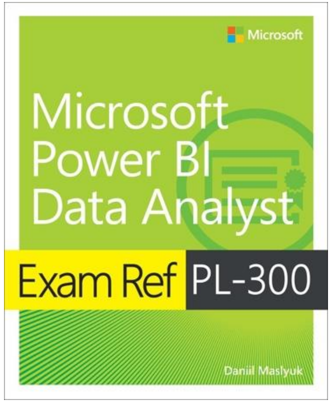

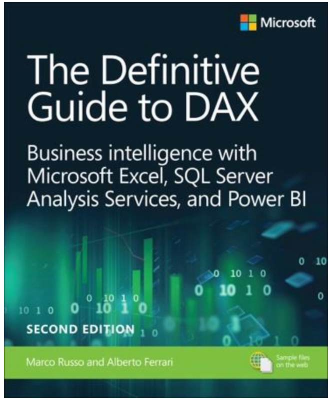

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Your Power BI experience

How long have you been using Power BI for?

- Less than 1 month
- 1 month ≤ X < 3 months
- 3 months ≤ X < 1 year
- 1 year ≤ X < 2 years
- 2 years ≤ X < 5 years
- 5 years or more

Course overview

- Lesson 1: Introduction
- Lesson 2: CALCULATE
- Lesson 3: Time Intelligence and other concepts

Lesson 1

Course introduction

- Introduction
- Calculated columns
- Measures
- Variables
- Calculated tables

Introduction

- DAX is one of the primary languages of Power BI
- DAX depends on your data model
- Knowing the theory is more important than for other programming languages

Calculated columns

- Calculations in a table
- Done row by row
- Same formula for each row

Measures

- Not everything can be calculated in columns
- Measures
    - Can aggregate columns and tables
    - Are calculated only when used in visuals or queries
    - Provide the consistency of logic and reusability
    - Can be formatted

Calculated columns vs. measures

- Create columns if you want to use the results
  - In a chart axis
  - Slicer
  - As a categorical filter
  - Etc.
- In most other cases, use measures

Variables

- Should be used for
  - Improving readability
  - Improving performance
  - Debugging
- Variables are immutable

Calculated tables

- Tables created by using DAX
- Can be dramatically faster than Power Query

Lesson 2

CALCULATE

- The use of CALCULATE
- Adding, removing, modifying filters with CALCULATE
- Using CALCULATE modifiers
- Evaluation contexts and context transition

CALCULATE

- The most important function in DAX
- Can modify the filter context by adding, removing, and updating filters
- Most commonly used with an expression and one or more filters

Modifying filters

- CALCULATE can
    - Add filters
    - Ignore filters
    - Override filters
- When CALCULATE filters conflict with visual filters, CALCULATE wins by default
- Simple filters are shorthand for table filters

CALCULATE modifiers

- USERELATIONSHIP
- CROSSFILTER
- KEEPFILTERS
- REMOVEFILTERS and the ALL family of functions

Evaluation contexts and context transition

- Filter context: comes from visual axes, slicers, report filters…
- Row context comes from the current row in a table
- CALCULATE in row context triggers context transition from row to filter context

Lesson 3

Time Intelligence and other concepts

- Time Intelligence
- SWITCH
- More table functions
- Window functions

Time intelligence

- You need a calendar table for time intelligence to work properly
- Avoid using the build-in calendar tables because they can't be used for more than one column and aren't flexible
- Some examples of time intelligence functions:
    - DATESYTD
    - DATEADD
    - DATESINPERIOD

SWITCH

- Useful to avoid nested IF statements
- Can be used in two ways: with
  - A value (like CASE in SQL)
  - TRUE (like CASE WHEN in SQL)
- Once the first condition is evaluated, no need to check it again

More table functions

- ADDCOLUMNS – adds columns to a table
- SELECTCOLUMNS – select columns from a table, keeping its grain
- SUMMARIZE – groups a table by columns
- DISTINCT and VALUES
- UNION
- Many more…

Window functions

- INDEX
- OFFSET
- RANK
- ROWNUMBER
- WINDOW
- Arguments
  - MATCHBY
  - ORDERBY
  - PARTITIONBY

Thank you!